

The Readout Shortcut: Positional Number Copying Dominates Arithmetic CoT Readout in Small Language Models

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Abstract

Chain-of-thought (CoT) prompting is necessary for arithmetic in small language models, yet shuffling its steps preserves most performance. What does CoT contribute if not logical sequencing? In three 1–3B instruction-tuned LMs on GSM8K, we isolate the answer-readout stage via prefix completion and identify a positional shortcut: the model copies whichever number occupies the trailing position before the answer delimiter, regardless of intermediate reasoning. Gold-answer presence accounts for 54–92 pp of accuracy (89–92% of each model’s teacher-forcing ceiling); even on incorrect items, the final answer matches the last CoT number 95–96% of the time. The copy channel takes precedence over retained-context completion: replacing the trailing number with a wrong value collapses accuracy to near-zero despite correct intermediates, yet removing it recovers 5–32 pp above that floor—even single-step arithmetic the model can otherwise perform is suppressed when a copyable number is present. Qwen and Llama copy novel distractors 87–95% of the time; Gemma gates selectively. Head-level ablation implicates architecture-specific head sets; the effect replicates on GSM-Symbolic. On non-arithmetic BBH tasks, shuffle retention drops sharply; at 7–8B, content-selective gating emerges. Step-level faithfulness evaluations risk conflating positional answer transport with genuine computation—a failure mode for CoT-based oversight.

1 Introduction

Chain-of-thought (CoT) prompting (Wei et al., 2022; Kojima et al., 2022) is necessary: removing it crashes accuracy (Lanham et al., 2023). Yet its step order barely matters: shuffling steps retains most performance (Madaan and Yazdanbakhsh, 2022; Wang et al., 2023a). This tension is well-documented but unexplained—*what* in CoT drives

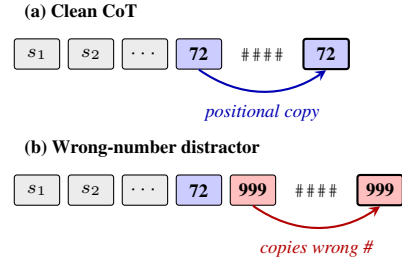


Figure 1: The *answer-context-gated positional readout*: the model reads whichever number appears in answer-relevant context at the trailing position before the #### delimiter. (a) When the correct answer is last, the readout yields the right output. (b) In Qwen/Llama (1–3B), injecting a wrong number in answer context displaces gold and is copied 87–95% of the time; Gemma instead shows stronger content gating ($P(\text{distractor}) = .12-.19$; §4.1).

the answer if not logical sequencing? Prior work established that CoT can be unfaithful—Turpin et al. (2023) showed biased demonstrations alter answers despite correct reasoning steps; Lanham et al. (2023) quantified unfaithfulness via early answering and paraphrase metrics; Arcuschin et al. (2025) observed similar patterns in naturalistic settings—and that filler tokens partially substitute for reasoning (Pfau et al., 2024); recent mechanistic studies have begun tracing information flow through CoT (Dutta et al., 2024) and probing which reasoning steps matter (Bogdan et al., 2025). These works document *that* readout can bypass reasoning but not *what specific signal* the readout uses, *where* in the network it operates, or *whether* it actively suppresses available computation. We provide all three.

We resolve this for 1–3B instruction-tuned models on arithmetic by identifying an *answer-context-gated positional readout* (Geirhos et al., 2020; McCoy et al., 2019): the model reads whichever number appears in answer-relevant context at the trailing position before the answer delimiter, largely in-

dependent of intermediate computation. The readout requires answer-relevant framing (Appendix U) and is correctness-independent in Qwen/Llama but content-gated in Gemma (Figure 1). On three architectures we show *what* the model reads, *why* shuffling preserves it (the answer token’s positional accessibility survives when logical order does not), and *where* it breaks (non-arithmetic tasks; content-selective retrieval at 7–8B). Gold-answer dependence is robust; the *content gate*—operationally, the degree to which the model rejects a novel distractor number at the trailing position, measured as $1 - P(\text{distractor})$ —varies from absent (Qwen: $\text{gate} \approx 0$) to strong (Gemma: $\text{gate} \approx .85$; 7–8B models: $\text{gate} \geq .70$; §7). Concurrent behavioral work (Chen et al., 2026) corroborates frontier-scale shuffle-tolerance; we contribute the mechanism, which requires open-weight access.

This characterization emerges from four converging lines of evidence on three architectures (Qwen2.5-1.5B-Instruct, Llama-3.2-1B-Instruct, Gemma-2-2B-it) on GSM8K (Cobbe et al., 2021); Gemma participates fully in the corruption decomposition, shuffle hierarchy, and head-level ablation, but its position/distractor results use a teacher-forcing-passing subset (Appendix T): (i) a corruption decomposition isolates gold-answer *presence* as the dominant factor (54–92 pp raw; 89–92% ceiling-corrected), and a seven-condition causal ladder shows the copy channel takes precedence over available retained-context completion (§3); (ii) the readout selects the trailing answer-context number with a sharp final-position jump (+20–31 pp) and novelty-permissive copying in Qwen/Llama (.87–.95 for novel numbers) vs. selective gating in Gemma (.12–.19; §4); (iii) head-level ablations reveal architecture-specific profiles from localized (Llama) to distributed (Qwen; §5); (iv) the shortcut is present in base weights, replicates on GSM-Symbolic, shifts toward selectivity at 7–8B, and collapses on non-arithmetic BBH (§6).

Scope and faithfulness. CoT use involves *rationale generation* (which may compute) and *answer readout* (which maps a completed prefix to the final token); our prefix-completion interventions isolate the second stage. Following Lanham et al. (2023), we treat a readout as *faithful* when the answer depends on intermediate computation ($\Delta_{\text{off-copy}}$) rather than surface features (Δ_{copy}). We make no claim about generation-time computation—only

that the readout does not faithfully use it when a trailing answer-context number is available. This undermines the premise of step-level CoT monitors (Lightman et al., 2023; Chen et al., 2025; Korbak et al., 2025): the readout takes precedence over available retained-context completion.

2 Experimental Setup

Models. Qwen2.5-1.5B-Instruct (Yang et al., 2024) (28L, 1536d, 12/2 GQA heads), Llama-3.2-1B-Instruct (Grattafiori et al., 2024) (16L, 2048d, 32/8 GQA heads), and Gemma-2-2B-it (Riviere et al., 2024) (26L, 2304d, 8/4 GQA heads). All use grouped-query attention (GQA; Ainslie et al., 2023) with per-head $\mathbf{o}_{\text{proj}}$ (attention output projection) columns, enabling per-head ablation. The three models span distinct tokenizer families (BPE variants for Qwen/Llama; SentencePiece unigram for Gemma), so convergent findings are not attributable to shared sub-word segmentation of numbers. Gemma appears in core experiments (corruption decomposition, shuffle hierarchy, head-level ablation, base-model probe); its position/distractor results (§4) show reduced teacher-forcing fidelity under prefix-structure perturbation ($\sim 60\%$ vs. $\sim 99\%$ on the corruption pipeline; Appendix T).

Task and protocol. GSM8K test set (Cobbe et al., 2021), first 500 problems. Baseline greedy CoT accuracy: Qwen 67.0%, Llama 45.6%, Gemma 66.2%. All experiments use teacher-forced prefix injection: a (possibly modified) CoT is injected as the beginning of the assistant turn using native chat templates, “#### ” is appended, and the model generates the answer greedily. This is the standard protocol for causal CoT analysis, isolating the readout from generation-time confounds (Lanham et al., 2023).¹

Statistical methodology. Wilson 95% CIs (Wilson, 1927) for proportions; McNemar’s exact test (McNemar, 1947) for paired contrasts; Holm–Bonferroni correction (Holm, 1979) within pre-declared confirmatory sub-families (Appendix S). Auxiliary analyses (induction overlap, mean-ablation, activation patching) are exploratory with uncorrected p -values.

Notation. $\Delta_{\text{copy}} = P_B - P_A$, $\Delta_{\text{off-copy}} = P_C - P_B$, and $P(\text{residual}) = P_A$ are *accuracy contributions*

¹Gemma-2-2B-it lacks native system-prompt support; we prepend the system message to the user turn.

(differences between conditional accuracies) that decompose P_C additively; we use Δ for these counterfactual contrasts and reserve $P(\cdot)$ for event proportions ($P(\text{distractor})$, $P(\text{gold})$) measured on a single output distribution. Causal-ladder conditions $D_{\text{rep}}/D_{\text{trunc}}/D_{\text{blank}}$ are defined in §3.3.

3 What Drives CoT Readout?

Among correctly-solved items, gold-answer presence accounts for 54–92 pp (89–92% of each model’s TF ceiling; §3.1); a causal ladder (§3.3) reveals that correct intermediates carry a latent signal (4–29 pp above no-CoT) that is masked when a trailing number is available. We condition on baseline-correct items; the free-generation diagnostic (§3.2) shows even incorrect answers match the last CoT number at .905–.957 across all three architectures.

3.1 Isolating the Gold-Presence Effect

Starting from correctly-solved problems (Qwen: $n=335$; Llama: $n=228$; Gemma: $n=331$), we construct three conditions by selectively corrupting numbers in the CoT prefix: **A (corrupt-all)** replaces all numbers including gold occurrences; **B (preserve-gold)** corrupts intermediates but leaves gold occurrences intact; **C (clean)** presents the original CoT. Corruption uses deterministic per-example seeding.²

Gold-presence ($\Delta_{\text{copy}} = P_B - P_A$) accounts for 54–92 pp raw across three architectures (Table 1). Gemma’s lower raw value reflects its reduced TF-fidelity ($P_C=.604$; the standard prefix format interacts with Gemma’s chat template). Ceiling-normalized copy strength (Δ_{copy}/P_C) converges at 88–92%, indicating a quantitatively comparable mechanism operating near each model’s achievable ceiling. Intermediate-step computation ($\Delta_{\text{off-copy}} = P_C - P_B$) is at noise floor for Qwen/Llama and modest for Gemma. Conditions A and B differ only in gold-answer presence, cleanly isolating this effect. Whether the copy is positional vs. content-selective is tested in §4.1.

3.2 Off-Ceiling Stress Test

To create headroom for measuring $\Delta_{\text{off-copy}}$, we swap in a weaker configuration—Qwen-base, 0-

²Items where corruption preserves the gold value (mostly single-digit golds in Gemma) are excluded from Condition A (Qwen/Llama: 5 each; Gemma: 51); including them yields $P_B - P_A \geq .789$ in all cases. Conditions B and C use the full sample.

Condition / Component	Qwen	Llama	Gemma
A: No gold in CoT (n_A)	.079 [.054, .113]	.117 [.081, .165]	.037 [.021, .063]
B: Gold present	1.00 [.989, 1.00]	.991 [.969, .998]	.580 [.526, .632]
C: Everything correct	1.00 [.989, 1.00]	.996 [.976, .999]	.604 [.551, .655]
$\Delta_{\text{copy}} = P_B - P_A$	.921 [.889, .946]	.875 [.828, .914]	.543 [.489, .597]
$\Delta_{\text{off-copy}} = P_C - P_B$	.000 [.000, .011]	.004 [.001, .024]	.024 [.009, .050]
$P(\text{residual}) = P_A$	.079 [.054, .113]	.117 [.081, .165]	.037 [.021, .063]
Δ_{copy}/P_C (ceiling-norm.)	.924	.879	.899

Table 1: Corruption decomposition (Qwen $n=335$, Llama $n=228$, Gemma $n=331$). Wilson 95% CIs for condition accuracies (rows A–C); paired bootstrap 95% CIs over items for derived differences (Δ_{copy} , $\Delta_{\text{off-copy}}$); $P(\text{residual})=P_A$ uses the Condition A interval directly. Gold-presence dominates (54–92 pp raw; 88–92% ceiling-normalized). Gemma’s lower raw Δ_{copy} reflects reduced teacher-forcing fidelity ($P_C=.604$) under the standard prefix format, not a weaker copy mechanism: ceiling-normalized copy strength converges across architectures (bottom row). Δ denotes accuracy *attributable to* each component (i.e., a difference between conditional accuracies), not an event probability.

shot ($P_B=0.735$, $n=200$)—with 26 pp of headroom, yet $\Delta_{\text{off-copy}}=0.020$ (bootstrap 95% CI $[-.015, .060]$). This is an order of magnitude below the .730 copy contribution. Correct intermediate steps add essentially no information to the readout.

If trailing-number alignment were a prefix-injection artifact, it should disappear under unconstrained decoding. It does not (Table 2): under free generation ($n=500$), all three models’ final answers match the last CoT number 96–97% of the time. Critically, on items the model gets *wrong*, the match rate is .905–.957—under a “compute-then-write” alternative, errors would distribute across non-trailing numbers, but they are pinned to the trailing position. Conditioning on whether gold occupies the last CoT position reveals a near-deterministic gate: accuracy is .991–.997 when gold is last vs. .004–.030 when it is not. This full-distribution analysis confirms the shortcut operates identically on items the model gets wrong, independent of any teacher-forcing intervention.

3.3 Copy Masks Retained-Context Computation

The decomposition above leaves a residual question: does $\Delta_{\text{off-copy}} \approx 0$ reflect genuine inability to complete from retained context, or does the copy channel *override* an available completion pathway? We disambiguate with four additional conditions that vary the trailing slot independently of intermediate-step correctness.

Metric	Qwen	Llama	Gemma
Baseline accuracy	.672	.448	.662
Final answer = last CoT num	.972	.974	.960
Acc gold is last	.997	.991	.994
Acc gold is <i>not</i> last	.030	.004	.023
Ans = last incorrect	.945	.957	.905

Table 2: Free-generation analysis ($n=500$, unconstrained greedy decoding, no prefix injection). Accuracy is near-perfectly predicted by whether gold occupies the last CoT position. Even on incorrect items, the answer reflexively matches the trailing number—the same positional-copy signature confirmed on all three architectures.

D-replace: correct intermediates, wrong trailing number. We hold intermediates correct (as in Condition B) but replace gold-answer occurrences with a wrong number, yielding Condition D_{rep} . Accuracy collapses to .076 (Qwen, $n=328$), .084 (Llama, $n=214$), .010 (Gemma, $n=312$)—within ≤ 1.1 pp of Condition A (.073, .079, .000), where intermediates are also corrupted (Table 3; pairwise McNemar $p > .01$; all nine §3.3 confirmatory contrasts yield raw $p < 10^{-8}$ and remain significant under Holm–Bonferroni correction at any family size up to the full 23-contrast set). The wrong number is copied with $P(\text{distractor}) = .924$ (Qwen), .911 (Llama), .574 (Gemma). Clean intermediates provide no benefit when a trailing distractor is present.

D-truncate: intermediates only, no trailing number. To probe whether this completion ability exists but is masked, D_{trunc} truncates the CoT before the first gold-answer occurrence (at the preceding sentence boundary, retaining $\sim 80\%$ of tokens), leaving only correct intermediate steps with no trailing number. Accuracy rises to .399 (Qwen, $n=308$), .183 (Llama, $n=208$), .058 (Gemma, $n=292$)—a +32/+10/+5 pp gain over D_{rep} : the same intermediates yield non-trivial accuracy once the trailing distractor is removed.³ A depth partition (Table 10) shows 74–77% of recovered items are 1-op reachable from the last visible intermediate; the shortcut thus suppresses even single-step arithmetic the model can otherwise perform.

Controls: No-CoT and D-blank. Two controls close remaining attack surfaces. **No-CoT** (direct

³Only 1–2% of D_{trunc} items have their new trailing number equal to gold (Qwen 5/308, Llama 4/208, Gemma 3/292); filtered rates (.389/.172/.055) are virtually identical to overall rates.

Model	Floor			Recompute		Ceiling	
	No-CoT	A	D_{rep}	D_{trunc}	D_{blank}	B	C
Qwen-1.5B	.106	.073	.076	.399	.461	1.000	1.000
Llama-1B	.019	.079	.084	.183	.197	.991	.991
Gemma-2B	.014	.000	.010	.058	.213	.571	.590

Copy-override gap ($D_{trunc} - D_{rep}$): Qwen +32pp, Llama +10pp, Gemma +5pp
Retained-context contribution ($D_{trunc} - \text{No-CoT}$): Qwen +29pp, Llama +16pp, Gemma +4pp
P(dist) in D_{rep} : Qwen .924, Llama .911, Gemma .574

Table 3: Causal ladder ($P(\text{gold})$, baseline-correct, per-experiment common-index subsets). Three regimes: **floor** (No-CoT, A, D_{rep})—trailing number absent or wrong, accuracy collapses; **recompute** (D_{trunc} , D_{blank})—removing distractor reveals available completion pathway; **ceiling** (B, C)—gold presence. Gemma B/C lower than Table 1 because truncation-length and leak filters disproportionately exclude Gemma items; per-condition n in Appendix S.

answer, no rationale prefix) yields .106 (Qwen), .019 (Llama), .014 (Gemma). The 29 pp gap $D_{trunc} - \text{No-CoT}$ on Qwen (McNemar $p < 10^{-12}$) shows the truncated intermediates carry genuine signal—the model is not re-solving from the question alone. D_{blank} appends a content-free closing sentence to the truncated prefix (restoring CoT format without a trailing number): .461, .197, .213. $D_{blank} \approx D_{trunc}$ on Qwen and Llama confirms that format completeness is not the confound; removing the trailing number, not the closing template, releases the available completion pathway.

Copy-over-recompute. Together: (a) retained-context completion exists ($D_{trunc} \gg \text{No-CoT}$, Qwen +29 pp, $p < 10^{-12}$); (b) the copy channel takes precedence ($D_{rep} \approx A$ despite identical intermediates); (c) the precedence is answer-context-gated ($D_{blank} \approx D_{trunc}$; cf. §4.2, Appendices U, E). “Override” is operational: the output matches the trailing number rather than the completion pathway’s prediction; we remain agnostic between active suppression and passive monopolization. The magnitude tracks copy strength: Qwen ($P(\text{distractor}) = .924$) shows the largest D_{trunc} gain (+32 pp); Gemma (.574) the smallest, with a +15 pp $D_{blank} - D_{trunc}$ gap suggesting additional format sensitivity. For CoT-based oversight, this is the load-bearing finding: evaluations that vary intermediates while preserving the trailing answer slot systematically underestimate available computation.

4 How Is the Answer Selected?

Three converging tests distinguish end-anchored copy from content-selective retrieval or generic recency (cf. Liu et al., 2024): both models copy wrong numbers placed at the end (§4.1); fixing

Cond.	Qwen ($n=335$)		Llama ($n=227$)	
	$P(\text{gold})$	$P(\text{dist})$	$P(\text{gold})$	$P(\text{dist})$
C0 (baseline)	1.00	—	.982	—
C0b (filler)	.782	—	.978	—
C1 (adjacent)	.051	.949 [.920,.968]	.053	.947 [.910,.970]
C2 (random)	.119	.881 [.841,.911]	.123	.872 [.823,.910]
C3 (gold dup.)	1.00	—	.996	—

Table 4: Distractor injection results. C0: baseline (no distractor); C0b: non-numeric filler; C1: adjacent wrong number (gold $\pm$ 1); C2: random wrong number; C3: gold duplicated (positive control). Both models copy a novel wrong number 87–95% of the time.

position recovers the shuffle drop with a sharp final-slot jump (§4.2); and a seven-level hierarchy shows performance tracks token accessibility (§4.3).

4.1 The Model Copies Wrong Numbers

If the readout is content-selective for gold, it should resist distractors (cf. Shi et al., 2023). Starting from the keep-end condition (accuracy $\approx$ 1.0), we append a wrong number between the answer step and the delimiter. We pre-register a decision rule: $P(\text{distractor}) > 0.70$ rejects gold-specific retrieval as the dominant readout (a content-selective readout predicts $P(\text{distractor})$ near 0; .70 ensures rejection requires substantial copy-dominance; observed values exceed this by 17–25 pp, so conclusions are insensitive to the exact threshold within [.50, .85]).⁴

Five conditions test adjacent (gold $\pm$ 1), random (same digit-count), and control distractors:

Both novel-distractor conditions exceed the .70 threshold by wide margins (Table 4; one-sided exact binomial: C1 $p < 10^{-15}$; C2 $p < 10^{-8}$). With novel numbers, the mechanism is predominantly not content-selective for gold in the 1–3B regime. A $\sim$ 7 pp C1–C2 gap suggests minor content sensitivity, well below .70. The asymmetric C0b sensitivity (Qwen -22 pp vs. Llama -0.4 pp from non-numeric filler) mirrors the topology contrast in §5: Qwen’s distributed copy circuit is destabilized by any trailing material, whereas Llama’s concentrated circuit implements a tighter numeric-target selector.

Novel-delimiter controls ($P(\text{distractor}) \geq .90$; Appendix E) rule out delimiter surface form. The “answer-context-gated” qualifier is established by a framing-dissociation experiment (Table 5): bare numbers at the trailing position yield low copying (.45/.26), while answer-relevant inline fram-

⁴Gemma’s Experiments 3–4 are excluded from headline comparisons; teacher-forcing fidelity falls below the .80 threshold (Appendix T). On the fidelity-passing subset ($n=196$), $P(\text{distractor})=.12-.19$.

Framing condition	Qwen		Llama	
	$P(\text{dist})$	$P(\text{gold})$	$P(\text{dist})$	$P(\text{gold})$
F1: Template (“the answer is X”)	.893	.107	.877	.123
F2: Bare number only	.448	.549	.260	.709
F3: Non-answer (“Note: X”)	.206	.788	.339	.652
F4: Inline (“should be X”)	.770	.110	.846	.137

Table 5: Framing-dissociation control ($n=335$ Qwen; $n=227$ Llama). F-codes denote framing conditions (distinct from distractor C-codes in Table 4). The readout is *answer-context-gated*: bare trailing numbers (F2) and non-answer framing (F3) fall well below the .70 threshold; answer-relevant inline text (F4) triggers copying comparably to standard templates (F1). This dissociates the mechanism from both raw positional recency and template-specific parsing.

ing (“actually it should be X”) recovers high copying (.77/.85)—same number, same position, but $P(\text{distractor})$ swings $\sim 2\times$ based purely on whether the surrounding text is answer-relevant. The readout is gated by answer-context framing, not raw position or template surface form.

Intermediate-result distractor. When the trailing number is an *intermediate computation result* (e.g., a subtotal from an earlier step)—already bound to a semantic role in the CoT—the models diverge ($n=200$). Qwen copies it 86% [.81,.90], consistent with novelty-permissive copying. Llama copies it only 25% [.20,.31], recovering the gold answer 74.5% of the time—evidence of a weak repetition-suppression filter layered on the dominant positional mechanism (25% remains far above the $\sim$ 1% no-distractor error floor, confirming positional copying still dominates). Architectures thus differ not in *whether* the shortcut is answer-context-gated but in *how sharply* the gate discriminates non-novel numerals: Qwen < Llama < Gemma.

4.2 Fixing Position Recovers the Shuffle Drop

If the readout is end-anchored, most of the shuffle drop should be recoverable by controlling where the answer step appears. We test four conditions: **ordered**, **full_shuffle**, **keep_end** (non-answer steps shuffled, answer fixed at end), and **move_front** (answer at front).

Keeping the answer step at the end recovers 99% (Qwen) / 93% (Llama) of the shuffle drop (Figure 2); ordered and keep_end differ by ≤ 1 pp (McNemar $p \geq .25$). The shallow-gradient-then-jump profile—+7/+10 pp over positions 0–75%, then +31/+20 pp at position 1.0—is inconsistent with smooth recency weighting.

A positional-encoding control (Appendix C)

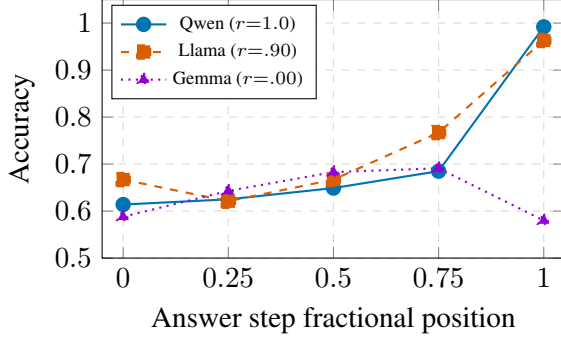


Figure 2: Answer-position curve (5-position sweep, $n=123\text{--}315$). Qwen/Llama show a shallow-gradient-then-jump profile: accuracy rises non-monotonically over 0–75% then jumps +31/+20 pp at position 1.0 (end). Gemma is flat (teacher-forcing fidelity issue; Appendix T). Discrete conditions: keep_end (pos=1.0) recovers ordered accuracy within ≤ 1 pp (McNemar $p \geq .25$).

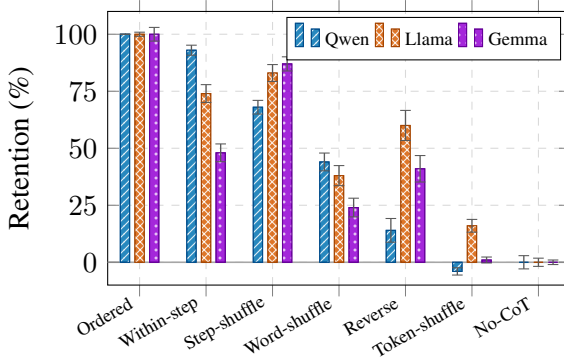


Figure 3: Shuffle hierarchy with bootstrap 95% CIs ($n=228\text{--}335$ items). Retention (no-CoT-anchored) = $(P_{\text{cond}} - P_{\text{no_cot}}) / (P_{\text{ord}} - P_{\text{no_cot}})$ anchors 0% at no-CoT and 100% at ordered. Step-level conditions preserve the answer token and retain 68–87%; token-level shuffling destroys token identity and collapses to no-CoT. Negative values indicate active interference.

rules out RoPE (Su et al., 2024) artifacts: monotonic position-ID stretching ($2.5\times$) causes no measurable accuracy loss ($n=66$), showing that position-encoding perturbation does not account for the shuffle drop.

4.3 Performance Tracks Token Accessibility

The binary keep_end/move_front contrast is coarse. A seven-level shuffle hierarchy decomposes the perturbation along structural granularity: step-level (step_shuffle, reverse_order), word-level (within_step, word_shuffle), and token-level (token_shuffle), plus ordered and no_cot anchors.

The hierarchy is monotone in answer-token accessibility (Figure 3): within each granular-

Condition	Qwen	Llama	Gemma
Baseline	1.000	.990	.697
Top-5 ablated	1.000 (–0pp)	.790 (–20pp)	.648 (–4.9pp)
Random-5 control	1.000 (–0pp)	.990 ($\Delta=0$ pp)	.709 (+1.2pp)
Max single-head	–0pp	–0pp	–19.4pp (L14H4)

Table 6: Zero-ablation sensitivity ($n=100$ Qwen/Llama; $n=165$ Gemma). Gemma’s lower baseline (.697) reflects its TF-fidelity-passing evaluation subset. Llama: 20pp drop from 5 heads, zero from random controls (permutation $p < .001$, $n=1,000$). Gemma: L14H4 single-head –19.4pp. Qwen: 0pp from top-5; cumulative ablation collapses at $K_{50}=17$, and mean-ablation shows real signal (–33.5 pp at $K=20$; §5.2).

ity level, conditions preserving the answer at the end (within_step) outperform those displacing it (reverse_order), so accessibility—not logical coherence—is the controlling variable. Self-generated CoT shuffle replicates the pattern on $N=2$ (Appendix I).

5 Architectural Variation in Head-Level Copy Sensitivity

Three interventions—zero-ablation, mean-ablation, and activation patching—converge on architecture-specific copy-sensitive head sets (Table 6). All three architectures participate in zero-ablation and mean-ablation; activation patching is reported for Qwen (the most distributed profile, where convergent evidence is most diagnostic). Gemma’s zero-ablation results ($K_{50}=5$, L14H4 single-head –19.4 pp) and induction-head overlap ($p < 0.01$) are fully included.

5.1 Zero-Ablation Reveals Three Distinct Sensitivity Profiles

Following Wu et al. (2024), we rank heads by mean attention mass from the last prefix token to gold-answer tokens ($n=100$ held-out), then zero their $\mathbf{o}_{\text{proj}}$ output columns on a disjoint evaluation set ($n=100$), ablating at query-head granularity under GQA. Specificity is tested with random-5 head sets (Llama: 20 layer-stratified, extended to $n=1,000$ permutation; Table 6).

Three profiles emerge (Table 6): Llama is *localized* (L10–L15; permutation $p < .001$, $n=1,000$); Gemma is *concentrated but compensable* (L14H4 dominant under zero-ablation, collectively load-bearing under mean-ablation; §5.2); Qwen is *distributed* ($K_{50}=17$, sharp phase transition at $K \approx 14\text{--}15$). After ablating Llama’s top-5 ($n=100$, acc.=.79; 21 failures), 0/21 are verbatim copies;

14/21 produce $2\times/3\times$ gold—a shift from copy-errors to recomputation-errors, indicating selectivity for the copy pathway rather than a general arithmetic deficit.

5.2 Converging Mechanistic Evidence

To address OOD concerns with zero-ablation (Zhang and Nanda, 2024), mean-ablation (replacing each head’s output with its mean over a held-out reference set) reproduces copy-specific top- K drops on all three architectures: Gemma -52 pp at $K=20$, Qwen -33.5 pp ($K_{50}=11$, retaining 43% of zero-ablation drop), Llama -6.5 pp peak. Single-head effects diverge between protocols (Gemma L14H4: -19.4 pp zero vs. -0.5 pp mean), but aggregate results converge. Activation patching on Qwen recovers 61% of the shuffle gap (Appendix Q). Content-tracking and induction-head analyses (Appendix P) corroborate a convergent-to-diffuse gradient: Gemma/Llama show significant induction/copy overlap ($p<0.01$); Qwen’s is near-chance. We apply Wu et al. (2024)’s attention-mass ranking outside its original long-context setting; relative to prior arithmetic-circuit work (Nikankin et al., 2025; Dziri et al., 2023; Stolfo et al., 2023; McDougall et al., 2023), we document a regime where the readout operates largely independently of internal arithmetic.

6 Generalization and Boundaries

We test memorization, instruction-tuning artifact, scale, and task specificity.

6.1 Novel-Instantiation Replication (GSM-Symbolic)

GSM-Symbolic (Mirzadeh et al., 2025) (main config, test split) regenerates GSM8K-style problems from symbolic templates with numeric values drawn outside the GSM8K distribution, eliminating verbatim overlap with pre-training data. Step-shuffle retains 73% (Qwen, $n=74$) / 81% (Llama, $n=86$)⁵—consistent with the GSM8K values (68% / 83%). On SVAMP (Patel et al., 2021), a structurally distinct arithmetic word-problem benchmark ($n=300$), the copy decomposition replicates: $\Delta_{\text{copy}} = 1.00$ (Qwen), .990 (Llama), .603 (Gemma; reduced by TF-fidelity; Appendix M). The copy phenomenon is not an artifact of memorization or GSM8K-specific distributional features.

⁵Qwen ceiling-corrected; Llama uncorrected (ordered acc. = 1.0).

Metric	1-3B	Qwen-7B	Llama-8B
Baseline accuracy	.670/.456	.894	.896
Δ_{copy}	.921/.875	.940 [.905, .970]	.798
$\Delta_{\text{off-copy}}$	.000/.004	.050	.196
$P(\text{distractor})$	.88-.95/.87-.95	.095	.045-.317

Table 7: Scale extension ($n=200$ evaluated; Llama-8B copy metrics computed on $n=163$ correct-baseline items). Both 7-8B models retain $\Delta_{\text{copy}} > .70$ while $P(\text{distractor})$ drops well below the .70 threshold. 1-3B column: Qwen-1.5B / Llama-1B. Qwen-7B $P(\text{distractor})$ uses the original single-condition protocol; Llama-8B range spans adjacent (C1) and random (C2) conditions.

6.2 Base-Model Origin

The copy mechanism is present in base-model weights (Qwen-base $\Delta_{\text{copy}}=.730$; Gemma-base .335; Llama-base .975 with 4-shot; Appendix O). Instruction-tuning amplifies copy strength (Gemma $1.6\times$ raw, .34 $\rightarrow$.54; ceiling-normalized .77 $\rightarrow$.90) but does not create the mechanism. Together with the novel-delimiter control (Appendix E) and GSM-Symbolic replication (§6.1), these results rule out IT-introduced format prior and delimiter-specific surface-form learning.

6.3 At 7-8B, Gold-Presence Persists but Copying Becomes Selective

Scaling up by $5\times-8\times$, we test Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct.⁶

Both 7-8B models retain gold-presence dependence ($\Delta_{\text{copy}} \geq .798$), but $P(\text{distractor})$ drops below .70 (Table 7): the novelty-permissive characterization does not hold at 7-8B. Llama-8B develops a substantial off-copy pathway ($\Delta_{\text{off-copy}}=.196$ vs. near-zero at 1B). With $N=2$ at this scale, these results are preliminary observations rather than a scaling claim.

An RL-distilled reasoning model (DeepSeek-R1-Distill-Qwen-1.5B; Guo et al., 2025) retains 96% ceiling-normalized copy-driven accuracy despite generating longer rationales (§7; Appendix K).

6.4 A Predicted Boundary: Shuffle Tolerance Disappears on Non-Arithmetic BBH

The copy account predicts shuffle tolerance should vanish when no copyable numeric answer occupies the trailing position. On BIG-Bench-Hard logical deduction (Suzgun et al., 2023; Srivastava et al., 2023) (3-way MCQ, letter answers), retention

⁶Gemma-2-9B-it excluded: pipeline debugging incomplete within submission timelines (Appendix T); not evidence about Gemma scaling.

Alternative explanation	Control / evidence
Gold-specific semantic retrieval	Wrong-number distractor (§4.1)
Smooth recency bias	Position sweep: final-slot jump (§4.2)
Answer-template parsing	Framing dissociation (Table 5)
Delimiter surface-form artifact	Novel-delimiter control (App. E)
GSM8K memorization	GSM-Symbolic replication (§6.1)
Instruction-tuning artifact	Base-model probe (§6.2)
Zero-ablation OOD	Mean-ablation + patching (§5.2)
Universal novelty-permissive copying	Gemma / 7–8B gating (§4.1, §6.3)
D_{trunc} is just re-solving	No-CoT floor $\ll D_{\text{trunc}}$ (§3.3)
Format-prior, not real copy	$D_{\text{blank}} \approx D_{\text{trunc}}$ (§3.3)

Table 8: Ten alternative explanations for the readout shortcut and the controls that address each. Every plausible confound is tested by at least one dedicated experiment or control condition.

drops to 44%/21% (Qwen/Llama)⁷—far below 68–87% on GSM8K. A second task (tracking shuffled objects) replicates: chance-corrected retention⁸ of 6.8% (Llama) / 45.5% (Qwen). The predicted collapse occurs on both tasks. These tasks differ from GSM8K along multiple axes (reasoning type, answer format, chance floor), so we characterize this as a boundary observation consistent with the shortcut’s preconditions rather than causal isolation of a single factor.

7 Discussion

The readout factors into two independently-varying components: robust gold-answer dependence (all five model×scale cells) and a content gate ranging from absent (Qwen) through partial (Llama-1B) and strong (Gemma-2B) to dominant at 7–8B. The causal ladder (§3.3) shows the copy channel takes precedence over available retained-context completion—predominantly single-step (74–77% of recovered items; Table 10)—making the shortcut’s dominance over even minimal arithmetic all the more diagnostic. The override magnitude (5–32 pp) tracks copy-channel strength. Because gold-presence dependence holds universally while the gate varies, order-insensitivity is a property of the readout’s invariant tier. Table 8 maps ten alternative explanations to the dedicated control that addresses each.

Implications for CoT-based oversight. Process-reward models (Lightman et al., 2023) and step-level monitors (Chen et al., 2025; Korbak et al., 2025) presuppose step quality is informative about the answer-producing computation. At 1–3B, the readout is largely independent of intermediate steps, so step-level rewards operate on a signal only

⁷Gemma’s BBH baseline is below chance (1/3), so retention is undefined.

⁸ $(p_{\text{shuffled}} - 1/3)/(p_{\text{ordered}} - 1/3)$; Appendix N.

weakly coupled to the answer-producing computation ($D_{\text{trunc}} - \text{No-CoT} = 4\text{--}29$ pp; §3.3). Note that process-reward models studied by Lightman et al. (2023) target much larger generators (GPT-4 scale); our concern applies specifically to 1–3B models, where content-selectivity is weakest. The override result (§3.3) sharpens this concern: a wrong answer-context number in the trailing slot is *worse* than no trailing number ($D_{\text{rep}} < D_{\text{trunc}}$), meaning a partially-corrupted CoT can be more misleading for oversight than a truncated one. As content-selectivity emerges at 7–8B (§6.3), monitorability should partially recover; output-side validators offer a practical mitigation for small-model deployments.

Reasoning-trained models. DeepSeek-R1-Distill-Qwen-1.5B (Guo et al., 2025) retains the shortcut despite longer rationales: ceiling-normalized $\Delta_{\text{copy}} = .928$ (96%; Appendix K). Its $P(\text{distractor}) = .706$ [.636, .768] ($n=180$) sits below Qwen/Llama (.87–.95), indicating that reasoning training introduces partial content selectivity without removing the positional copy mechanism. The persistence claim rests on Δ_{copy} ; the lower distractor acceptance suggests the content gate is partially engaged even at 1.5B when reasoning-trained.

Practical recommendations. For 1–3B deployments: output-side answer verification should be preferred over step-level process rewards, since the readout’s near-independence from intermediate steps (§3.3) leaves step-level signals only weakly coupled to the answer-producing computation; stripping answer-template framing may partially disrupt the shortcut (Table 5); and monitoring whether the answer matches the last CoT number provides a simple copy-dominance diagnostic.

8 Conclusion

In three 1–3B instruction-tuned LMs, the CoT readout is dominated by a positional shortcut: the model reads the trailing number in answer-relevant context. A causal ladder reveals that intermediate steps carry exploitable retained-context signal (4–29 pp above no-CoT) that is masked when a trailing number is available—the copy channel takes precedence over retained-context completion. Gold-answer dependence is robust; content-gating varies by architecture and scale.

Limitations

Scope. Primary evidence covers 1–3B instruction-tuned models on arithmetic (GSM8K, GSM-Symbolic, SVAMP [Appendix M]). The copy-detection paradigm requires a numeric gold answer at an identifiable trailing position, which structurally restricts the primary scope to arithmetic; we test the predicted boundary on two non-numeric BBH tasks (§6.4). Scale extension to 7–8B is $N=2$ (Qwen, Llama); Gemma-2-9B-it was deferred for pipeline reasons (see §6.3 footnote), not because preliminary data was unfavorable. Non-arithmetic coverage is $N=2$ BBH tasks (logical deduction, tracking shuffled objects), where shuffle retention collapses to 7–46% chance-corrected (§6.4); extending the diagnostic to non-numeric reasoning would require a different operationalization of answer-position copying and remains open. Reasoning-trained model coverage is $N=1$ (DeepSeek-R1-Distill-Qwen-1.5B); whether the shortcut persists across reasoning-distillation methods or base architectures remains open.

Architecture-specific caveats. Content-blindness varies across architectures (§4.1). Gemma’s Exp 3–4 are excluded due to teacher-forcing fidelity drops (Appendix T).

Methodology. Causal contrasts use prefix completion (Lanham et al., 2023), the field-standard protocol for isolating readout from generation-time confounds. Three ecological checks corroborate the readout-shortcut pattern under unconstrained decoding: (i) free-generation answers match the last CoT number $\sim 97\%$ overall and .905–.957 on incorrect items (Appendix G, $N=3$); (ii) self-generated CoT shuffle replicates the step>word>token hierarchy (Appendix I, $N=2$ step-only, $N=1$ full); (iii) gold-position gating in free generation (.991–.997 vs. .004–.030) mirrors the prefix-completion decomposition. Distractor and position effects under unconstrained generation remain future work. D_{trunc} retains $\sim 80\%$ of the CoT and provides a *lower bound* on retained-context contribution; the retained prefix may end with a near-gold number (e.g., a penultimate-step result), so the D_{trunc} –No-CoT gap reflects any exploitation of retained context, including one-step completion as well as multi-step retained-context derivation (a depth partition in Table 10 quantifies this: 74–77% of items are 1-op reachable). Head-

level ablation identifies sensitivity profiles, not complete circuits; behavioral evidence (§3–3.3) is primary for the copy-dominance claim, with mechanistic localization (§5) as supporting evidence.

Ethics Statement

We restrict experiments to controlled prefix interventions on open-weight models (Qwen2.5 [Apache-2.0], Llama-3 [Llama Community License], Gemma-2 [Gemma Terms of Use]) and open datasets (GSM8K [MIT], GSM-Symbolic [CC-BY-NC-ND-4.0], BBH [MIT]). Compute for reported experiments: ~ 40 GPU-hours; including pilot runs and debugging iterations, total project compute is ~ 60 – 80 GPU-hours (Appendix B). Practitioners deploying small-model CoT monitors should not treat rationales as load-bearing evidence of the answer. Our findings expose a vulnerability that could be exploited: an adversary aware of the trailing-number copy channel could construct CoTs whose intermediate steps appear to reason correctly while routing the final answer through the positional shortcut, evading step-level monitors. This risk is most acute for 1–3B models in cost-constrained deployments. We mitigate disclosure risk by restricting demonstrations to open-weight models in the 1–3B regime, reporting concrete defenses (§7), and showing that content-selectivity recovers at 7–8B. We believe public disclosure is net-positive for safety: it equips evaluators with a diagnostic and discourages premature reliance on small-model CoT as evidence of computation.

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A Experimental Details

Table 9 summarizes the design of all experiments reported in the paper.

Number corruption (§3.1). Intermediate numbers corrupted with deterministic per-example seeding: outer seed via `hashlib.sha256` of problem index and condition string, inner number perturbation via `hashlib.md5` of the number string and seed offset. Corruption magnitude: $\max(1, \lfloor 0.3 \times |v| \rfloor)$, direction chosen pseudorandomly. Gold answer occurrences detected via regex with digit boundaries. Items where the gold value persists in the corrupted text (due to accidental collision or small-number preservation) are excluded from Condition A (see footnote in §3.1).

Causal ladder and controls (§3.3). D_{rep} : starting from Condition B (correct intermediates, gold present), all gold-answer occurrences are replaced with a single deterministic wrong number per item (same corruption seed for all gold positions). D_{trunc} : the CoT is truncated at the sentence boundary preceding the first gold-answer occurrence; sentence boundaries are detected by the regex `(?<=[.!?\\n])\\s+`, with fallback to the last newline and then to a character-level cut when no boundary precedes gold. Items retaining $< 30\%$ of text or leaking gold into the truncated prefix are excluded. Truncated prefix length: mean 78.8%/79.7%/78.9% of original CoT tokens (Qwen/Llama/Gemma). D_{blank} : the D_{trunc} prefix is appended with “The answer is determined by the steps above.\\n\\n#### ”. No-CoT: prefix is “#### ” only. Common sample sizes: D_{rep} —Qwen $n=328$, Llama $n=214$, Gemma $n=312$; $D_{\text{trunc}}/D_{\text{blank}}/\text{No-CoT}$ —Qwen $n=308\text{--}310$, Llama $n=208\text{--}213$, Gemma $n=292\text{--}296$ (slight variation due to truncation-length filter). D_{trunc} trailing-number partition: items where the new trailing number = gold are rare (Qwen 5/308, Llama 4/208, Gemma 3/292); excluding them yields .389, .172, .055—essentially identical to overall D_{trunc} accuracy.

D_{trunc} depth partition. To assess whether D_{trunc} accuracy reflects primarily one-step completion (the gold answer is reachable from the last retained intermediate in a single arithmetic operation) or multi-step retained-context derivation, we classify each D_{trunc} correct item by depth. An item is *1-op reachable* if the gold answer can be obtained from any pair of numbers in the last retained sentence

via a single arithmetic operation ($+$, $-$, $\times$, $\div$); otherwise it is *multi-step*.

Distractor control (§4.1). Distractors have same digit-count as gold. Adjacent: $\text{gold} \pm 1$. Random: uniform from same-magnitude range, rejection-sampled to avoid gold. Templates: “Therefore, the answer is {X}.” etc. Qwen: $n=335$; Llama: $n=227$.

Answer-position (§4.2). Answer step identified as last step containing gold answer with digit-boundary regex. Steps split by paragraph breaks; sentence-level fallback. Qwen: $n=335$; Llama: $n=227$ (1 excluded due to < 3 steps).

Shuffle hierarchy (§4.3). Seven conditions: ordered, `within_step_shuffle`, `step_shuffle`, `word_shuffle`, `reverse_order`, `token_shuffle`, `no_cot`. Stochastic conditions: 5 seeds; deterministic: seed 0 only. For stochastic conditions, each item’s accuracy is the mean over its 5 seeds; reported accuracies and bootstrap CIs treat per-item means as the sample units. Sequence length filter: 1024 tokens. Problems with < 2 steps are excluded.

Head-level ablation (§5.1). Requires `attn_implementation="eager"`. Phase 1/Phase 2 use disjoint held-out splits (50/50). Attention mass: mean over answer-token positions. Ablation zeroes $\mathbf{o}_{\text{proj}}$ columns $[h \cdot d_h : (h+1) \cdot d_h]$. Random-5 control: 20 layer-stratified sets for Llama; single non-copy sets for Qwen and Gemma. Sequence length filter: 1536 tokens.

Top attention-mass heads. Llama: L11H14 (.383), L12H13 (.348), L10H13 (.293), L11H20 (.268), L15H12 (.252). Gemma: L18H6 (.146), L16H6 (.120), L17H4 (.114), L17H3 (.107), L16H4 (.101).

Generation. Greedy decoding, max 32 new tokens for prefix-continuation experiments; max 512 for free-generation (§G). Hardware: NVIDIA A10G (24GB), `bfloat16`. EOS: `<|eot_id|>` (Llama), `<|im_end|>` (Qwen), `<end_of_turn>` (Gemma).

B Reproducibility

All experiments: greedy decoding (temperature=0), deterministic per-example seeding (outer seed via `hashlib.sha256` of problem index and condition string; inner number perturbation via `hashlib.md5`). All

Experiment	Section	n (per model)	Baseline-correct?	TF-passing?	Paired?	Seeds	Primary test
Corruption decomposition (A/B/C)	§3.1	228–335	Yes	Yes (P_C)	Yes	1	McNemar
$D_{\text{rep}}/D_{\text{trunc}}/D_{\text{blank}}/\text{No-CoT}$	§3.3	208–328	Yes	Yes (P_C)	Yes	1	McNemar
Shuffle hierarchy (7 cond.)	§4.3	228–335	Yes	Yes (ordered)	Yes	5 (stoch.)	Bootstrap CI
Answer position (5 positions)	§4.2	123–315	Yes	Yes (pos=1.0)	Yes	5	Spearman
Distractor injection (C1/C2)	§4.1	227–335	Yes	Yes (baseline)	Yes	1	Binomial vs. .70
Head-level ablation	§5.1	100–165	Yes	Yes	No (disjoint)	1	Permutation
GSM-Symbolic	§6.1	300–500	No	—	No	1	Wilson CI
SVAMP	App. M	300	No	—	No	1	Wilson CI
BBH (2 tasks)	§6.4	64–106	Yes	Yes (ordered)	Yes	1	McNemar
Scale (7–8B)	§6.3	163–200	Yes/No	Yes	Yes	1	Wilson CI
Novel-delimiter distractor	App. E	200	Yes	Yes	Yes	1	Wilson CI
Bare-number control	App. U	227–335	Yes	Yes	Yes	1	Wilson CI
Base model	App. O	200	No	—	No	1	Wilson CI
Reasoning model	App. K	200	No	—	No	1	Wilson CI
Free-generation	App. G	500	No	—	No	1	Proportion
Self-gen shuffle	App. I	500	Yes (self)	—	Yes	5	Bootstrap CI

Table 9: Summary of all experiments. *Baseline-correct*: items conditioned on the model answering correctly under Condition C (clean prefix). *TF-passing*: teacher-forcing fidelity check applied (metric in parentheses). *Paired*: same items across conditions (enables McNemar). *Seeds*: number of stochastic seeds for randomized conditions.

Model	Category	n	Acc	No-CoT	Δ
Qwen	1-op reachable	237	.451	.106	+34.5 pp
Qwen	Multi-step	71	.225	.106	+11.9 pp
Llama	1-op reachable	158	.228	.019	+20.9 pp
Llama	Multi-step	50	.040	.019	+2.1 pp
Gemma	1-op reachable	216	.065	.014	+5.1 pp
Gemma	Multi-step	76	.039	.014	+2.5 pp

Table 10: D_{trunc} depth partition. 1-op reachable items (74–77% of total) account for most D_{trunc} accuracy; multi-step items (23–26%) still exceed the No-CoT floor on all models, indicating some genuine multi-step derivation, though at reduced magnitude.

stochastic operations (corruption magnitude, distractor generation, step reordering) derive from these per-item seeds, making results bitwise-identical regardless of global seed choice. Per-example outcome records are released in the result JSONs for reproducibility. Hardware: bfloat16 on NVIDIA A10G GPUs. Software: PyTorch 2.5.1–2.6.0, HuggingFace Transformers 5.7.0–5.8.1, Python 3.11–3.12. Compute for reported experiments: ~ 40 GPU-hours; including pilot runs and debugging iterations, total project compute is ~ 60 – 80 GPU-hours. Code is available at <https://github.com/mlzoo/cot-readout-shortcut>.

C Positional Encoding Control

Monotonic position-ID stretching ($2.5\times$) on ordered CoT causes no measurable accuracy loss (0/66 errors), showing that position-encoding perturbation alone does not reproduce the shuffle effect. A RoPE (Su et al., 2024) 2×2 factorial (Table 11) crossing content order with position-ID assignment shows: the content-shuffle main effect

(-31.3 pp) dominates the position-encoding channel.

	Uniform pos.	OOD pos.
Ordered content	1.000	.357
Shuffled content	.687	.223
Δ (shuffle effect)	-31.3 pp	-13.3 pp

Table 11: RoPE 2×2 factorial ($n=300$, Qwen). “OOD pos.”=monotonically increasing position IDs with random gaps (Uniform $\{1, \dots, 5\}$). Content-shuffle dominates; OOD positions cause large drops but are confounded by the gap distribution being OOD for the model’s RoPE.

D Answer Position Curve

For problems with ≥ 5 steps, we place the answer step at fractional positions $\{0, 0.25, 0.5, 0.75, 1.0\}$; non-answer steps shuffled (5 seeds per position).

Position	Qwen ($n=300$)	Llama ($n=123$)	Gemma ($n=315$)
0.00 (front)	.614 [.558, .667]	.667 [.580, .744]	.588 [.533, .641]
0.25	.625 [.569, .678]	.621 [.533, .702]	.643 [.590, .695]
0.50	.649 [.593, .701]	.667 [.580, .744]	.683 [.629, .731]
0.75	.685 [.630, .735]	.767 [.685, .833]	.691 [.638, .739]
1.00 (end)	.992 [.974, .998]	.963 [.913, .985]	.580 [.525, .633]
Spearman r	1.00 ($p=.017$)	.90 ($p=.083$)	.00 ($p=1.0$)

Table 12: Answer-position curve. Spearman p -values are exact two-sided permutation tests ($n=5$ positions; with only 5 data points, the per-position CIs and effect sizes carry the primary statistical weight). Qwen/Llama: shallow-gradient-then-jump recency. Gemma: no monotonic effect (consistent with the teacher-forcing fidelity issue noted in §4).

E Novel-Delimiter Distractor Control

Three delimiters designed to be absent from pretraining ($\gg$ RESULT:, ##FINAL##, [ANSWER]) tested with distractor injection ($n=200$ per delimiter).

Delimiter	Qwen ($n=200$)		Llama ($n=200$)	
	$P(\text{dist})$	$P(\text{gold})$	$P(\text{dist})$	$P(\text{gold})$
$\gg$ RESULT:	.995 [.972,.999]	.005	.975 [.943,.989]	.020
##FINAL##	.995 [.972,.999]	.005	.905 [.856,.938]	.055
[ANSWER]	.995 [.972,.999]	.005	1.00 [.981,1.00]	.000
#### (standard)	1.00	.000	.890	.110

Table 13: Novel-delimiter control ($n=200$ per condition). All conditions exceed .70 threshold by large margins. The format-prior account predicts a drop with unseen delimiters; none observed. Standard-delimiter row uses the same $n=200$ subsample for comparability.

F Delimiter Robustness

Three familiar delimiters tested on Qwen ($n=200$ each): ####, Answer:, The answer is . All yield $P_B > .95$ and $P(\text{distractor}) = .96\text{--}.965$ with overlapping CIs (pairwise McNemar $p > .7$).

G Free-Generation Consistency

Under unconstrained decoding ($n=500$ per model), all three architectures’ answers match the last number in their own CoT 96–97% of the time (Table 14). Critically, $P(\text{answer}=\text{last num} \mid \text{incorrect}) = .945$ (Qwen) / .957 (Llama) / .905 (Gemma)—even wrong answers match the last CoT number. Conditioning on whether the gold answer occupies the last CoT position reveals a near-deterministic gate: accuracy is .991–.997 when gold is last vs. .004–.030 when it is not. This full-distribution analysis extends the baseline-correct decomposition (§3.1) to the entire sample without baseline-correct conditioning. The .905–.957 match rate on *incorrect* items is particularly diagnostic: under a “compute-then-write” alternative where the trailing number reflects the model’s computation, errors would distribute across non-trailing numbers; instead, even wrong answers reflexively match the trailing token—the same positional-copy signature observed under teacher-forced corruption, now confirmed to operate during the model’s own unconstrained generation.

H Base Model Few-Shot Control

Llama-base 4-shot ICL resolves format collapse ($P_C : .065 \rightarrow .985$) and yields $\Delta_{\text{copy}}=.975$,

Metric	Qwen	Llama	Gemma
Baseline accuracy	.672	.448	.662
Final answer = last CoT num	.972	.974	.960
Gold is last CoT num	.664	.450	.658
Acc gold is last	.997	.991	.994
Acc gold is <i>not</i> last	.030	.004	.023
Ans = last incorrect	.945	.957	.905

Table 14: Full-distribution free-generation analysis ($n=500$ per model, greedy decoding). The readout shortcut is not an artifact of teacher-forcing or baseline-correct conditioning: across all items including incorrect ones, accuracy is near-perfectly predicted by whether the gold answer is the last CoT number. The positional-copy mechanism operates identically during the model’s own unconstrained generation.

$P(\text{distractor})=.890$. The 0-shot null is consistent with format collapse rather than mechanism absence; however, 4-shot ICL simultaneously provides format scaffolding and demonstrations of the copy pattern, so we cannot fully separate latent-mechanism elicitation from ICL-induced pattern matching.

I Self-Generated CoT Shuffle

To test whether the shuffle hierarchy persists under the model’s own CoTs (rather than teacher-forced human-written CoTs), we collect Llama-3.2-1B-Instruct’s free-generation CoTs on GSM8K ($n=500$, 5 seeds), filter to the correctly-solved subset, and apply step-shuffle, word shuffle, and token-shuffle to the model’s own outputs. The modified self-generated CoT is then injected as a teacher-forced prefix and the model generates the answer.

Llama replicates the hierarchy: step-shuffle retains 82.1%, word shuffle 77.0%, and token-shuffle 14.6%—matching the teacher-forced ordering (step > word > token). Qwen’s step-shuffle retention is 67.1%—closely matching its teacher-forced value (68.4%), confirming step-shuffle ecological validity on both primary models ($N=2$); the full hierarchy (step > word > token) is shown for Llama ($N=1$). However, Qwen’s word-level conditions are confounded by a format artifact: Qwen’s self-generated CoTs use a markdown-numbered-list format, and 79.6% of step-shuffle errors are list-continuation outputs (the model generates “1. ...” instead of an answer). This format dominance is itself consistent with the paper’s “format-driven copy” thesis—the model perseverates on list continuation rather than answer extraction when the list order is disrupted—but it prevents clean mea-

surement of word-level conditions for Qwen. We therefore report the full hierarchy for Llama only, with Qwen’s step-shuffle confirmation noted above.

J Scale Extension: Full Results

See main text Table 7. Additional detail: Qwen-7B $P_B=.950$ [.910, .973], $P_A=.010$ [.003, .036], $P_C=1.00$ [.981, 1.00] ($n=200$). Llama-8B $\Delta_{\text{copy}}=.798$, $\Delta_{\text{off-copy}}=.196$, $P(\text{residual})=.006$ ($n=163$ correct of 200 evaluated). The substantial off-copy pathway in Llama-8B (19.6 pp vs. near-zero at 1B) suggests the larger model partially recovers correct answers from intermediate reasoning steps.

K Reasoning Model

DeepSeek-R1-Distill-Qwen-1.5B (Guo et al., 2025): $P_C=.967$, $\Delta_{\text{copy}}=.928$, $\Delta_{\text{off-copy}}=.033$, $P(\text{residual})=.006$, $P(\text{distractor})=.706$ ($n=180$). Ceiling-normalized (Δ_{copy}/P_C): 96.0% copy-driven vs. 92.6% for standard Qwen ($P_C=1.00$). RL reasoning training does not develop substantial off-copy pathways.

L Cumulative-K Ablation Sweep

M Cross-Task: SVAMP

$\Delta_{\text{copy}} = 1.00$ (Qwen) / .990 (Llama) / .603 (Gemma) on SVAMP (Patel et al., 2021) ($n=300$). Gemma’s lower SVAMP value reflects the same teacher-forcing fidelity issue as in GSM-Symbolic: $P_C=.665$ on SVAMP (vs. ~ 1.0 for Qwen/Llama), so the Gemma SVAMP decomposition is unreliable. A similar issue affects Gemma’s GSM-Symbolic ceiling-corrected step-shuffle retention, which yields an uninterpretable 117% due to the low teacher-forcing fidelity. Position sensitivity is attenuated on Qwen/Llama (keep_end – full_shuffle = +9 pp vs. +18–30 pp on GSM8K), plausibly due to SVAMP’s redundant lexical answer cues; Gemma’s keep_end – full_shuffle is negative (–8.5 pp), consistent with the teacher-forcing fidelity issue described above.

N BBH Full Results

Additional: Qwen free-gen baseline .344; Llama free-gen baseline .260. Llama’s below-chance shuffled accuracy (CI [.043, .187], excluding .333) indicates active interference rather than mere signal absence.

O Base Model Full Results

P Supplementary Head-Level Analyses

Scope. The analyses below characterize head-level sensitivity, not a verified computational sub-graph; full circuit identification is left to follow-up work. An activation-patching screen (§Q) provides in-distribution causal evidence complementing the zero-ablation sensitivity analysis.

Induction-score protocol. Following Olsson et al. (2022): per head, the *prefix-matching score* is the average attention from position $K+i$ back to position $i+1$ on $N=200$ random-token repeated sequences $[r_1 \dots r_K][r_1 \dots r_K]$ ($K=50$, vocabulary restricted to digits, operators, and common math tokens); the *copying score* is the OV-circuit direct-logit attribution of each head’s contribution to the gold (first-half) token at second-half positions. The “induction score” cited in-text is the product of prefix-matching and copying scores; per-head score pairs and the previous-token-head check on the top-5 causal heads are released with the code.

Content-tracking: Under word-shuffle, 95.8% of Qwen’s 336 heads show content Jaccard > position Jaccard (Monte Carlo null: 0/336 heads; mean null Jaccards $\approx 0.8\%$). Ordering-sensitivity: Gini = 0.738 ($p < 0.001$), but layer-matched random control yields permutation $p=0.38$ —shuffled CoT is generically fragile under any ablation. Induction overlap (hypergeometric test on top-20 sets): Gemma L18H6 = top induction head (score 0.44), 7/20 overlap ($p=9.2 \times 10^{-4}$); Llama $\rho=0.248$, 5/20 overlap ($p=5.7 \times 10^{-4}$); Qwen $\rho=0.046$ (induction vs. patching-screen ranking), 2/20 overlap ($p=0.34$, near-chance), full-rank induction-vs.-ordered-attention-mass $\rho=0.254$ ($p=2.3 \times 10^{-6}$)—a gradient from convergent (Gemma) through partial (Llama) to diffuse (Qwen). RoPE 2×2 factorial: see Appendix C.

Q Activation Patching Screen

For each example, we cache per-head $\mathbf{o}_{\text{proj}}$ inputs from an ordered run and a shuffled run, then replace the shuffled run’s head slice with the ordered slice and measure logit recovery on the gold-answer token (Wang et al., 2023b). This is an in-distribution activation replacement—both source and target are real model activations—and is therefore immune to the OOD critique that applies to zero-ablation.

On Qwen ($n=150$ validation, disjoint from the $n=34$ screening set used for head selection with

K	0	1	5	8	10	12	14	15	16	17	18	19	20
Acc	.995	.965	.985	.965	.935	.895	.855	.675	.635	.565	.460	.310	.220

Table 15: Cumulative-K zero-ablation on Qwen ($n=200$). Phase transition at $K=14-15$; $K_{50}=17$.

Condition	Qwen ($n=86$)		Llama ($n=65$)	
	Acc	Ret.	Acc	Ret.
Ordered	.895	100%	.431	100%
Shuffled	.395	44%	.092	21%
Chance	.333	—	.333	—

Table 16: BBH logical deduction. Retention = shuffled/ordered (simple ratio); Figure 3 uses the no-CoT-anchored formula; §6.4 tracking task uses chance-corrected retention to account for the MCQ floor. Under either metric, BBH retention (44%/21%) is far below GSM8K (68–87%). Llama drops below chance—shuffled CoT actively interferes.

Condition	Qwen ($n=106$)		Llama ($n=64$)	
	Acc	CI	Acc	CI
Ordered	.887	[.812, .934]	.719	[.599, .814]
Shuffled	.585	[.490, .674]	.359	[.253, .482]
Chance	.333	—	.333	—
Retention (simple)	66.0%		50.0%	
Retention (chance-corr.)	45.5%		6.8%	
McNemar p	1.9×10^{-7}		1.5×10^{-5}	

Table 17: BBH tracking shuffled objects (3-way MCQ). Chance-corrected retention = $(p_{\text{shuffled}} - 1/3)/(p_{\text{ordered}} - 1/3)$. Both models show highly significant shuffle effects; Llama’s chance-corrected retention (6.8%) approaches the MCQ floor.

$|\Delta \text{LD}| > 0.3$ nat filter): patching the top-20 heads (by recovery ratio) restores accuracy from .693 (shuffled baseline) to .880 (group-patch), recovering 61% of the shuffle gap. The top individual head by validation accuracy (L0H8) achieves .813 alone (vs. .693 shuffled); the top screening-rank head (L14H8) achieves .787. Split-half Jaccard for the top-20 set is $.496 \pm .074$ (50 random splits), indicating stable head selection. The Gini coefficient across all 336 heads is .738 (0/1,000 permutations exceed observed; $p < .001$), indicating that the recovery effect is concentrated in a small head subset rather than broadly distributed—a random-20 set drawn from this distribution would recover substantially less than the top-20’s 61%. Of the top-20 patching heads, 6 overlap with the top-20 attention-mass heads (30%), confirming that attention mass and causal recovery are correlated but non-identical rankings.

Metric	Qwen (0-shot)	Llama (0-shot)	Gemma (0-shot)
P_C	.755	.065	.435
P_B	.735	.055	.340
P_A	.005	.015	.005
Δ_{copy}	.730	.040	.335
$P(\text{distractor})$	.715	.055	.300

Table 18: Base model decomposition ($n=200$ each). Llama’s near-zero results reflect format collapse, not absence of mechanism (see Appendix H).

R Paraphrased CoT Control

Replacing numbers with English words causes -2.4% ($p=.013$, $n=335$); operator paraphrasing: 0% effect. Errors concentrate on items where the final answer is spelled out—only the answer-slot token form matters.

S Statistical Methodology Details

We use the simplest test appropriate to each contrast’s design (paired/unpaired, one/two-sided) and correct within each pre-declared confirmatory sub-family. The sub-families contain 23 contrasts in total: **Exp 1** A vs. B ($\times 2$ models, 2 McNemar); **Exp 3** ordered vs. full_shuffle, ordered vs. keep_end, keep_end vs. full_shuffle, keep_end vs. move_front ($\times 2$ models, 8 McNemar); **Exp 4** C1 and C2 vs. $P=0.70$ ($\times 2$ models, 4 one-sided binomial); §3.3 causal ladder D_{rep} vs. A, D_{trunc} vs. No-CoT, D_{blank} vs. D_{trunc} ($\times 3$ models, 9 McNemar). Sensitivity: doubling one-sided p -values before correction leaves all significant. Wilson CIs are used for single-condition proportions; derived differences ($\Delta_{\text{copy}} = P_B - P_A$, $\Delta_{\text{off-copy}} = P_C - P_B$) use paired bootstrap over items (10,000 resamples), which correctly accounts for within-item dependence across conditions. Bootstrap and Wilson intervals agree within ± 1 pp. Mean-ablation and activation-patching results are exploratory and not part of the confirmatory contrast family. The intermediate-result distractor experiment (Qwen $P=.860$; Llama $P=.250$) is reported in §4.1 as evidence for architecture-dependent content gating.

T Teacher-Forcing Fidelity Diagnostic

Teacher-forcing fidelity measures whether a model produces the expected continuation when given its

own correct CoT as a prefix. We report fidelity for each experiment $\times$ model combination to make exclusion decisions transparent. The fidelity metric is experiment-specific: for Exp 1, P_C (accuracy under Condition C); for Exp 2, ordered-condition accuracy; for Exp 3, accuracy at position = 1.0 (unperturbed answer placement); for Exp 4, baseline distractor-free accuracy.

	Exp 1 (P_C)	Exp 2 (ordered)	Exp 3 (pos = 1.0)	Exp 4 (baseline)
Qwen	1.00	1.000	.992	1.00
Llama	.996	.996	.963	.982
Gemma	.997	.931	.580	.586
Threshold	.80 — cells below are excluded from analysis			

Table 19: Teacher-forcing fidelity per experiment $\times$ model. Gemma’s fidelity drops below the .80 threshold on Exp 3 (answer position) and Exp 4 (distractor injection), where structural perturbation causes the model to deviate from the teacher-forced prefix. These cells are excluded from the main analysis (§4); a TF-passing subset analysis (below) shows the core patterns replicate when conditioning on fidelity-passing items. All other cells exceed .93.

The .80 threshold is conservative: the gap between passing cells ($\geq .931$) and failing cells ($\leq .60$) is large, so any threshold in $[\cdot60, \cdot93]$ yields identical inclusion decisions—the specific value is immaterial. The Exp 1 column (P_C) serves as the TF-fidelity diagnostic for the entire corruption pipeline (Conditions A, B, C, D_{rep} , D_{trunc} , D_{blank} , No-CoT), all of which share the prefix-completion infrastructure. D_{rep} modifies only the trailing number (a textually mild perturbation); the model’s acceptance of the corruption is captured directly by $P(\text{distractor})$. D_{trunc} removes the trailing tail by design, so TF fidelity in the “expected continuation” sense is not defined for that condition.

TF-passing subset analysis. As a robustness check, we condition Gemma’s Exp 3 and Exp 4 on the subset of items where teacher-forcing fidelity passes (i.e., the model produces a parseable answer under the unperturbed condition). On this subset ($n=196$), Exp 3 patterns replicate: ordered .990, full-shuffle .847, keep_end .949, move_front .735 (recovery = 71%). The 5-position answer-placement curve yields Spearman $r=1.0$ ($n=187$ TF-strict), restoring the monotonic recency gradient absent in the full sample. For Exp 4, TF-passing $P(\text{distractor})$ is .194 $[\cdot145, \cdot255]$ (C1) and .117 $[\cdot079, \cdot170]$ (C2)—well below the .70 thresh-

old, confirming that Gemma’s Exp 3 exclusion reflects a teacher-forcing artifact rather than a genuine mechanistic divergence from Qwen and Llama.

U Bare-Number Distractor Control

All main-experiment distractors are wrapped in answer-template sentences (“Therefore, the answer is {X}.” etc.). A reviewer might attribute the high $P(\text{distractor})$ to template parsing rather than positional copying. We test four conditions that vary answer-relevant framing while keeping the distractor number identical ($n=335$ Qwen; $n=227$ Llama):

Condition	Qwen		Llama	
	$P(\text{dist})$	$P(\text{gold})$	$P(\text{dist})$	$P(\text{gold})$
F1: Template	.893	.107	.877	.123
F2: Bare number	.448	.549	.260	.709
F3: Non-answer (“Note: X”)	.206	.788	.339	.652
F4: Inline (“...should be X”)	.770	.110	.846	.137

Table 20: Bare-number distractor control (F-codes as in Table 5). F1: standard template (“Therefore, the answer is X”); F2: bare number only; F3: non-answer framing (“Note: X”); F4: natural inline answer-revision (“But wait, actually it should be X”). The readout requires answer-relevant framing ($F2/F3 \ll .70$) but is not template-specific ($F4 \geq .70$).

Bare numbers (F2) and non-answer-context numbers (F3) fall well below the .70 threshold on both models, showing the readout is not purely positional—a trailing number without answer-relevant framing is insufficient. However, natural inline answer-revision text (F4) triggers copying at rates comparable to the standard template (.770/.846 vs. .893/.877), showing the mechanism is not template-specific. The readout is *answer-context-gated*: it requires some form of answer-relevant framing at the trailing position, but is indifferent to the specific surface form of that framing and—within answer context—remains indifferent to whether the number is correct.

V Alternative Explanations Summary

W Qualitative Examples

We illustrate the copy-dominance phenomenon with representative cases from Qwen2.5-1.5B-Instruct on GSM8K.

Example 1: Distractor copied (Condition C1). The gold answer is 72. A wrong-number distractor

Alternative explanation	Control / evidence
Gold-specific semantic retrieval	Wrong-number distractor (§4.1)
Smooth recency bias	Position sweep: final-slot jump (§4.2)
Answer-template parsing	Bare-number control (§U)
Delimiter surface-form artifact	Novel-delimiter control (§E)
GSM8K memorization	GSM-Symbolic replication (§6.1)
Instruction-tuning artifact	Base-model probe (§6.2)
Zero-ablation OOD	Mean-ablation + patching (§5.2)
Universal novelty-permissive copying	Gemma / 7-8B gating (§4.1, §6.3)
D_{trunc} is just re-solving	No-CoT floor $\ll D_{\text{trunc}}$ (§3.3)
Format-prior, not real copy	$D_{\text{blank}} \approx D_{\text{trunc}}$ on Qwen/Llama; Gemma + 15 pp gap noted (§3.3)

Table 21: Alternative explanations for the readout short-cut and the controls that address each. Each alternative is tested by at least one dedicated experiment or control condition.

sentence (“Therefore, the answer is 45.”) is appended at the trailing position after the correct CoT. The model outputs **45** (the distractor), ignoring the correct computation in the prefix. This occurs in 87% of Qwen trials.

Example 2: D_{trunc} succeeds. Same problem; CoT is truncated before the final step (removing the trailing gold-answer number). The model now outputs **72** (correct), recovering the answer from the retained penultimate-step context. This demonstrates the retained-context computation (29 pp above No-CoT for Qwen) that is *masked* when a trailing number is present.

Example 3: Distractor resisted (Gemma content gate). Same Condition C1 setup on Gemma-2-2B-it. The model outputs **72** (gold answer), rejecting the novel distractor. Gemma’s content gate ($1 - P(\text{distractor}) \approx .85$) prevents the copy channel from accepting semantically implausible trailing numbers.